



AI BUG TRACKER — SMART DEBUG ASSISTANT

Time Limit: 4 Hours

1. Project Overview

Build an intelligent web application that accepts and analyzes error logs.

The system must:

- Detect and redact sensitive information before sending to AI.
- Use AI to explain the root cause, categorize the issue, suggest fixes, and estimate severity.
- Implement deduplication and caching to avoid reprocessing similar logs.

Evaluates full-stack development, AI integration, regex logic, caching efficiency, and security best practices.

2. Technical Stack

Frontend: React.js or similar, Tailwind CSS or Material-UI, Axios or Fetch API

Backend: Node.js (Express/Fastify), Python (FastAPI/Flask), or Java (Spring Boot)

Database: SQLite or PostgreSQL

AI: OpenAI GPT-4 or GPT-4o

3. Core Functional Requirements

A. Log Upload and Parsing

- Accept .log, .txt, or .json files (max 5MB)
- Support drag-and-drop or manual upload
- Display file metadata and show log content in a scrollable panel

B. Sensitive Data Redaction

- Redact file paths, IPs, usernames, API keys, URLs, timestamps
- Send only sanitized logs to AI

4. Intelligent Log Handling

A. Duplicate Log Prevention

- Generate SHA-256 hash; detect duplicates and prompt reuse.

B. Smart AI Caching

- Reuse cached AI results for similar logs.

C. Dashboard & Upload History

- Display filename, upload time, file size, severity, and analysis status.

5. Security and Validation

- Validate file type and size
- Sanitize inputs
- Store API keys securely

6. AI Explanation and Output

AI output must include: Issue Type, Root Cause, Suggested Fix, Severity Rating.
Display results on the dashboard for each upload.

7. Evaluation Criteria

- Functionality: Proper parsing, redaction, and AI integration
- Deduplication: Detect identical or similar logs
- Caching: Efficient reuse of AI explanations
- Code Quality: Modular and documented
- UI/UX: Clean and responsive

8. Deliverables

- Complete runnable source code
- README with setup instructions
- Application demo.

9. Notes

- Use your own OpenAI key
- Focus on functionality and clarity
- Document incomplete parts
- Justify design decisions

All the best!